

Statistics is the science of collecting, analyzing, and interpreting data. It helps us understand patterns and make decisions.

The mean is the average of a set of values. To calculate it, sum all values and divide by the number of values.

The median is the middle number in an ordered data set. If there is an even number of values, the median is the average of the two middle numbers.

The mode is the value that appears most frequently in a dataset.

Standard deviation measures the spread of data from the mean. A low standard deviation means the data is close to the mean.

Probability is the measure of the likelihood that an event will occur.

In a normal distribution, the mean, median, and mode are all equal.

A histogram is a graphical representation of the distribution of numerical data.

Correlation measures the relationship between two variables. A positive correlation means they increase together.

A sample is a subset of a population, used to estimate characteristics of the whole group.

Hypothesis testing is a method used to determine if there is enough evidence to support a specific claim about a population.